A Speculative $(3 + 3)$ Spacetime Model: Wick-Rotated Mass Generation, Phase-Space Entropy Gradients, and Quadrupolar Gravitational Radiation

A Personal Thought Experiment — Not a Peer-Reviewed Physical Theory

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DISCLAIMER. This document is a **personal thought experiment and speculative idea exploration**, written by a physics enthusiast with AI assistance. It is **not** a peer-reviewed publication, and it does **not** present an established or proven physical theory. The "derivations" herein are heuristic analogies rather than rigorous proofs, and the model is known to contain unresolved mathematical and physical problems, several of which are listed in Section 6. Readers should treat this text as a record of creative exploration, not as established science.

Abstract

Standard General Relativity formulates cosmological dynamics within a (3+1)-dimensional pseudo-Riemannian manifold under the invariance of the speed of light (c). In this speculative note, we explore a $(3 + 3)$-dimensional spacetime concept comprising three spatial coordinates ($\mathbf{r} \in \mathbb{R}^3$) and three temporal dimensions ($\mathbf{t} = (t_1, t_2, t_3) \in \mathbb{R}^3$). Borrowing $Sp(2, \mathbb{R}) \times U(1)$-type gauge constraints from Two-Time Physics, we sketch a possible route toward taming the negative-norm ghost states typical of multi-time formulations (this route is *not* proven here). Applying a Wick rotation to an internal temporal axis ($t_2 \to -i\tau_2$), we *interpret* rest mass m_0 as an intrinsic oscillation energy ($E_2 = m_0c^2$) of localized "system photons" (γ_{sys}); we emphasize that this identification is definitional rather than dynamical. We further sketch a thermodynamic picture of the arrow of time via phase-space entropy gradients, and restate — without new derivation — the standard distinction between static $1/r^2$ gravitational fields and quadrupolar gravitational radiation within this picture. Schematic (non-rigorous) experimental directions are outlined, and known open problems of the framework are listed explicitly.

1 Introduction and Extended Gauge Formalism

The mathematical incompatibility between Quantum Mechanics (QM) and General Relativity (GR) stems from the ontological nature of measurement, spatial geometry, and intrinsic mass. While GR treats spacetime as a classical $(3+1)$-dimensional continuum, QM relies on non-unitary wavefunction collapse and scalar rest masses defined in an abstract Hilbert space.

As a speculative construction, we consider a six-dimensional pseudo-Riemannian manifold $\mathcal{M}^6$ equipped with dimensionless scale factors (α_2, α_3):

$$ds^2 = dx^2 + dy^2 + dz^2 - c^2 dt_1^2 - \alpha_2^2 c^2 dt_2^2 - \alpha_3^2 c^2 dt_3^2 = g_{MN} dX^M dX^N \tag{1}$$

where $X^M = (\mathbf{r}, ct_1, \alpha_2 ct_2, \alpha_3 ct_3)$. In macroscopic domains we assume $\alpha_2, \alpha_3 \sim \ell_P/\lambda_C \ll 1$ (where ℓ_P is the Planck length and λ_C the Compton wavelength), intended to yield a low-energy reduction to the classical $(3+1)$-dimensional Minkowski metric. (See Section 6 for a known tension between this assumption and the mass identification of Section 3.)

1.1 Axiom of Light-Speed Invariance and Null Geodesics

In $\mathcal{M}^6$, the **Constancy of the Speed of Light** is postulated as an unbroken invariant. For any fundamental field propagating along a Null Geodesic ($ds^2 = 0$), the geometric line element yields:

$$dr^2 = c^2 dt_1^2 + \alpha_2^2 c^2 dt_2^2 + \alpha_3^2 c^2 dt_3^2 = c^2 \|\mathbf{d}\tau_{\text{temporal}}\|^2 \tag{2}$$

Equation (2) is intended to bound spatial displacement by the norm of the temporal vector, motivating (but not proving) the absence of tachyonic signals and closed timelike curves (CTCs).

1.2 Toward Ghost Elimination via $Sp(2,\mathbb{R}) \times U(1)$ Gauge Symmetry

Multi-time geometries naturally introduce negative-norm states (ghosts) due to additional time-like metric signatures. Inspired by Two-Time Physics [2], we sketch a set of constraints over the phase-space operators $(\hat{X}^M, \hat{P}^M)$:

$$\hat{H}_0 |\Psi_{\text{phys}}\rangle = \eta_{MN} \hat{P}^M \hat{P}^N |\Psi_{\text{phys}}\rangle = 0 \tag{3}$$

$$\hat{H}_2 |\Psi_{\text{phys}}\rangle = (\hat{P}_2^2 + \hat{X}_2^2)|\Psi_{\text{phys}}\rangle = 0, \quad \hat{H}_3 |\Psi_{\text{phys}}\rangle = (\hat{P}_3^2 + \hat{X}_3^2)|\Psi_{\text{phys}}\rangle = 0 \tag{4}$$

Caveat: as written, the operators $\hat{P}_i^2 + \hat{X}_i^2$ are positive-definite, so Equation (4) admits no nonzero solutions in a standard Hilbert space; a consistent treatment would require a full BRST / gauge-fixing analysis, which is beyond this note. Ghost elimination in this framework therefore remains an *open problem*, not a result.

2 Phase-Space Entropy Densities and Symmetry Breaking

While the kinematic metric g_{MN} exhibits $SO(3,3)$ rotational symmetry, macroscopic reality displays an irreversible temporal arrow. We sketch a thermodynamic picture of this asymmetry via microstate phase-space ensemble dynamics.

2.1 Phase-Space Density Projection Vector

Let $\rho(\mathbf{q}, \mathbf{p}; \mathbf{t})$ represent the microstate phase-space distribution function evolving across $\mathcal{M}^6$. The Gibbs entropy $S_{\text{ensemble}} = -k_B \int \rho \ln \rho \, d\Omega$ is conjectured to generate a non-symmetric thermodynamic flux vector $\mathcal{A}_{\mathbf{t}}$ in temporal space:

$$\mathcal{A}_{\mathbf{t}} = \left(\frac{\delta S_{\text{ensemble}}}{\delta t_1}, \frac{\delta S_{\text{ensemble}}}{\delta t_2}, \frac{\delta S_{\text{ensemble}}}{\delta t_3} \right) \tag{5}$$

Cosmological low-entropy initial conditions would then establish a dominant macroscopic entropy production along t_1:

$$\frac{\delta S_{\text{ensemble}}}{\delta t_1} \gg 0, \quad \left\langle \frac{\delta S_{\text{ensemble}}}{\delta t_2} \right\rangle_{\text{macro}} = 0, \quad \left\langle \frac{\delta S_{\text{ensemble}}}{\delta t_3} \right\rangle_{\text{macro}} = 0 \tag{6}$$

By Landauer's Principle [5], memory formation and clock advancement require local dissipation ($\Delta S > 0$). The intuition is that observers would be confined to advance along $\mathcal{A}_{\mathbf{t}} \parallel \mathbf{e}_{t1}$, effectively reducing $SO(3,3)$ to a macroscopic $SO(3)_{\text{space}} \times SO(2)_{\text{temporal}}$ structure. No dynamical equations for $S(\mathbf{r}, \mathbf{t})$ are provided here; this remains a qualitative picture.

2.2 Microscopic Phase Fluctuations and Zero-Point Energy

At micro-scales ($\sim \ell_P$), local ensemble variance is conjectured to persist:

$$\left\langle (\Delta S_{t_2})^2 \right\rangle_{\text{micro}} = \sigma_S^2 \neq 0 \tag{7}$$

We speculate that such micro-temporal noise might relate to Heisenberg's Uncertainty Principle ($\Delta E_2 \Delta t_2 \geq \frac{\hbar}{2}$) and vacuum zero-point oscillations; no quantitative connection is derived.

3 Wick-Rotated Rest Mass Interpretation

To address the metric sign inconsistency in the multi-time mass picture, we apply a local Euclideanization (Wick rotation) to the internal quantum temporal coordinate $t_2 \to -i\tau_2$. **Note**: after this rotation τ_2 enters the metric with a spacelike sign, so the framework at this point is effectively $(4+2)$-dimensional rather than $(3+3)$; see Section 6.

3.1 Heuristic Identification $E_2 = m_0 c^2$

Under Wick rotation, the internal timelike micro-interval transforms as $-c^2 dt_2^2 \to +c^2 d\tau_2^2$. The generalized 6D momentum-energy relation for fundamentally massless fields ($ds^2 = 0$) takes the form:

$$\mathbf{p}^2 - \frac{1}{c^2} E_1^2 + \frac{1}{\alpha_2^2 c^2} E_2^2 - \frac{1}{\alpha_3^2 c^2} E_3^2 = 0 \tag{8}$$

For an unmonitored particle ($E_3 = 0$) at spatial rest ($\mathbf{p} = \mathbf{0}$), Equation (8) simplifies to:

$$E_1^2 = \frac{E_2^2}{\alpha_2^2} \implies E_1 = E_2 = m_0 c^2 \quad (\text{for } \alpha_2 = 1) \tag{9}$$

Interpretation (not derivation): comparing with $E^2 = \mathbf{p}^2 c^2 + m_0^2 c^4$ suggests reading rest mass m_0 as the energy of a system photon (γ_{sys}) localized in 3D space and oscillating along the internal coordinate τ_2, echoing De Broglie's internal frequency $\omega_0 = m_0 c^2/\hbar$. We stress that this is a relabeling: no dynamics is provided that determines E_2, so the identification has no predictive content by itself. The condition $\alpha_2 = 1$ used here also conflicts with the macroscopic assumption $\alpha_2 \ll 1$ of Section 1.

3.2 Tripartite Classification of Electromagnetic Fields

We classify photon field interactions according to their functional roles across $\mathcal{M}^6$. This classification mirrors the standard system / apparatus / environment decomposition of decoherence theory [3]:

Table 1: Tripartite Functional Classification of Photons in the $(3+3)$ Picture

Photon Class	**Temporal Axis**	**Phase Entropy Flux**	**Physical Function**
System Light (γ_{sys})	Quantum Phase (τ_2)	$\frac{\delta S}{\delta \tau_2} = 0$	Internal oscillation ($E_2 = m_0 c$
Measurement Light (γ_{meas})	Constraint Axis (t_3)	$\Delta S_{\text{info}} > 0$	Null-geodesic junction; state lo
Environmental Light (γ_{env})	Cosmic Time (t_1)	$\frac{\delta S}{\delta t_1} \gg 0$	Phase randomization; thermod

4 Decoherence, Static Gravitational Fields, and Quadrupolar Waves

Under Coulomb gauge, the effective potential $\Psi(\mathbf{r}, \mathbf{t})$ is postulated to follow the generalized 6D wave equation:

$$\left(\nabla_{\mathbf{r}}^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t_1^2} + \frac{1}{\alpha_2^2 c^2}\frac{\partial^2}{\partial \tau_2^2} - \frac{1}{\alpha_3^2 c^2}\frac{\partial^2}{\partial t_3^2}\right)\Psi(\mathbf{r}, \mathbf{t}) = 0 \tag{10}$$

4.1 Quantum Superposition and Natural Decoherence

For an isolated system (γ_{sys} alone), wave propagation behind a double slit yields a coherent superposition $\Psi_{\text{total}} = \psi_A + \psi_B$. Recorded intensity on a detector array at t_1 is given by:

$$I(\mathbf{r}) \propto |\Psi_{\text{total}}|^2 = |\psi_A|^2 + |\psi_B|^2 + 2\text{Re}\left(\psi_A^* \psi_B\right) \tag{11}$$

Scattering with environmental photons (γ_{env}) introduces stochastic phase shifts $\phi_{env}(\mathbf{t})$. Performing an ensemble average over the microscopic temporal dimensions eliminates non-diagonal interference:

$$\left\langle e^{i\phi_{env}(\mathbf{t})}\right\rangle_{\tau_2, t_3} = \int e^{i\phi_{env}(\mathbf{t})} d\tau_2 dt_3 \to 0 \implies \rho(\mathbf{r}, \mathbf{r}') \to |\psi_A|^2 + |\psi_B|^2 \tag{12}$$

This phase-averaging step is mathematically the standard environmental decoherence mechanism [3], here re-expressed in the multi-time notation; the extra temporal dimensions do not produce results beyond standard decoherence theory in this calculation.

4.2 Static Gravitational Fields vs. Dynamic Quadrupolar Waves

We restate — without new derivation — the standard distinction between static gravitational fields and radiative waves, expressed within the $(3+3)$ picture:

1. **Static Gravitational Field**: A system photon γ_{sys} oscillating along τ_2 is interpreted as rest mass $E_2 = m_0 c^2$. A stationary mass distribution creates a spherically symmetric, static gravitational flux $\mathbf{g}$ across 3D space, satisfying Gauss's Law:

$$\nabla_{\mathbf{r}} \cdot \mathbf{g} = -4\pi G \rho_{\text{mass}}(\mathbf{r}, \tau_2) \implies \mathbf{g}(\mathbf{r}) = -\frac{GM}{r^2}\hat{\mathbf{r}} \tag{13}$$

   By Birkhoff's Theorem, isolated spherically symmetric masses yield static attraction without monopolar radiation. (This is standard Newtonian gravity / GR, not a result of the $(3+3)$ model.)

2. **Dynamic Quadrupolar Waves**: Gravitational radiation $h_{\mu\nu}$ is emitted when mass distributions undergo non-axisymmetric accelerated motion, propagating transverse quadrupolar tensor waves at speed c:

$$\Box \bar{h}_{\mu\nu} = -\frac{16\pi G}{c^4} T_{\mu\nu}^{(\text{quadrupole})} \tag{14}$$

   (This is standard linearized GR, consistent with LIGO observations; it is restated here for context.)

5 Schematic Experimental Directions (Non-Rigorous)

The following are concept-level sketches, *not* predictions rigorously derived from the model.

5.1 Attosecond Phase Corrections (Sketch)

If an internal axis τ_2 exists, ultra-fast attosecond ionization experiments ($\Delta t \sim 10^{-18}$ s) might exhibit an additional phase term of the schematic form:

$$\delta\Phi_{\text{corr}} \sim \alpha_2 \left(\frac{m_0 c^2}{\hbar}\right) \Delta\tau_2 \tag{15}$$

Here $\Delta\tau_2$ lacks an operational definition and would need to be specified before any comparison with experiment is possible.

5.2 Fundamental Qubit Dephasing Limits (Sketch; formula needs rebuilding)

We conjecture that even fully isolated qubits ($\gamma_{env} \to 0$) retain a minimal dephasing floor set by microscopic fluctuations σ_S^2 along τ_2. The previously proposed bound $\Gamma_{\text{dephasing}} \geq \sigma_S^2 k_B/\hbar$ is dimensionally inconsistent (it does not have units of a rate) and is retained here only as a directional idea pending a correct formulation.

6 Limitations and Open Problems

For intellectual honesty, we list the currently known unresolved problems of this framework:

1. **Wick rotation vs. "three times"**: after $t_2 \to -i\tau_2$, the coordinate τ_2 is effectively spacelike, making the framework $(4+2)$-dimensional and contradicting the central "three temporal dimensions" claim.

2. **$E_2 = m_0c^2$ is definitional**: no Lagrangian or dynamics determines E_2; the identification is a relabeling of Einstein's energy relation and carries no predictive content.

3. **Scale-factor inconsistency**: Section 1 assumes $\alpha_2 \ll 1$ while Section 3 requires $\alpha_2 = 1$.

4. **Constraint equations as written have no nonzero solutions**: $\hat{P}_i^2 + \hat{X}_i^2$ is positive-definite; a proper BRST / gauge-fixing analysis is absent, so ghost elimination is unproven.

5. **Ultrahyperbolic Cauchy problem**: the loss of well-posed initial-value evolution in multi-time settings [1] is not actually resolved here.

6. **Measurement does not require photon scattering**: which-path experiments (quantum eraser, cavity QED) demonstrate interference loss without momentum-disturbing photons, undermining the physical motivation for the t_3 "measurement light" axis.

7. **Overlap with standard decoherence theory**: the calculable content of the tripartite photon scheme reproduces standard decoherence theory; the extra temporal dimensions add interpretation, not new results.

8. **No new gravitational results**: Section 4.2 restates Newtonian gravity and linearized GR rather than deriving them from the 6D framework.

7 Conclusion

This note has recorded a speculative (3+3) spacetime picture connecting quantum superposition, an oscillatory interpretation of rest mass, a thermodynamic account of the arrow of time, and a restatement of static versus radiative gravity. The framework contains suggestive echoes of real physics — Two-Time Physics [2], dimensionality analyses [1], decoherence theory [3], and De Broglie's internal clock — but, as detailed in Section 6, it remains a thought experiment with substantial unresolved problems rather than a consistent physical theory. We share it in the spirit of open exploration and welcome criticism.

References

[1] M. Tegmark, "On the dimensionality of spacetime," *Class. Quantum Grav.*, vol. 14, no. 4, pp. L69–L75, 1997.

[2] I. Bars, "Two-Time Physics," *Phys. Rev. D*, vol. 74, p. 085019, 2006.

[3] W. H. Zurek, "Decoherence, einselection, and the quantum origins of the classical," *Rev. Mod. Phys.*, vol. 75, pp. 715–775, 2003.

[4] J. A. Wheeler, "The 'past' and the 'delayed-choice' double-slit experiment," in *Mathematical Foundations of Quantum Theory*, Academic Press, pp. 9–48, 1978.

[5] R. Landauer, "Irreversibility and Heat Generation in the Computing Process," *IBM J. Res. Dev.*, vol. 5, no. 3, pp. 183–191, 1961.